Neural Network Handoff Algorithm in a Joint Terrestrial-HAPS Cellular System

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ABSTRACT

Handoff algorithm is used in wireless cellular systems to decide when and to which base station (BS) to handoff in order that the services can be continued uninterrupted. In this paper, we propose a handoff algorithm based on the neural network in a joint system of terrestrial and high altitude platform station (HAPS) cellular systems. As a revolutionary wireless system, HAPS can supply services for uncovered area improving total capacity of service-limited area by a terrestrial BS. Radial-Basis function (RBF) network is used for making a handoff decision to the chosen neighbor BS. The neural inputs consist of the averaged signal strength received from the serving and nearby BSs, user directions estimated by the MUSIC algorithm on an antenna array, and traffic intensities. Positioning a mobile station (MS) is obtained by apply the timing advance (TA) concept. Performance comparisons among the presented method and using the backpropagation (BP) neural network and the conventional Hysteresis method are given in forms of (1) handoff rate, blocking rate, dropping rate at the acceptable grade of service and (2) the difference between the signal power radiated by the serving BS and the minimum required received signal power before call dropping for both Hata model and Rayleigh fading.

Keywords: Handoff, Radial-basis function networks, Backpropagation networks, MUSIC, Timing advance, Rayleigh fading.

1. INTRODUCTION

In mobile communications, the continuity of communication without terminating an ongoing call or blocking new calls is very crucial to enhance high quality of cellular services. Handoff algorithm makes it possible to maintain link quality. In Hysteresis method [1], handoff occurs when the difference of signal strength received from target and current BSs is higher than Hysteresis level. Because of fading effect, the difference can be fluctuated for brief periods of time which results in unnecessary handoffs. Such back and forth handoff is known as the ping-pong effect. In addition to network resource waste, calls might be terminated if decision delay is long due to the high Hysteresis level.

Recently, neural networks have been utilized to improve handoff algorithms due to its ability to handle large data in fast processing. Adaptive parameters such as user speeds, received signal strengths for pattern classification provide a multiple of criteria handoff algorithm [2]. Neural network is trained to predict a user's transfer probabilities which represent the user movements [3]. A technique to recognize signal patterns of a MS using probabilistic neural network is introduced in Rayleigh fading channel [4]. Using statistical pattern recognition of signal strength [5-6] can improve the efficiency of handoff algorithms.

In this paper, we present an effective handoff algorithm based on radial basis function (RBF) networks [7] in a joint terrestrial-HAPS cellular system [8]. Figure 1 shows the conceptual model of the joint cellular system. The inputs of neural network depend on signal strengths, mobile directions and traffic intensities, which are used to make a handoff decision to the chosen adjacent cell or HAPS. We use an antenna array with the MUSIC [9] (MUltiple SIgnal Classification) to estimate directions of arrival of MS signals. The signal strengths of mobiles are computed in the Hata model and Rayleigh fading [10]. HAPS cellular system can be considered as a complementary to the terrestrial cellular system, to improve and expand the coverage services. As shown in Fig. 1, HAPS can supply services to the mobile having weak signal from the serving terrestrial BS influenced by shadowing, turning corner as well as being outside the terrestrial coverage. The timing advance (TA) concept [11-12] in GSM system and the power of mobile signal are used for checking if the mobile are in an obstacle area.

This paper is organized as follow. Section 2 reviews the Hysteresis method. Handoff decision based on RBF network is presented in section 3 and based on the backpropagation network is next presented in section 4. The MUSIC algorithm is explained in section 5. The concept of timing advance is in section 6. The handoff algorithm and HAPS improving the call maintenance are presented in section 7. Channel propagations for both Hata and Rayleigh are modeled in section 8. Section 9 illustrates the simulation results followed by conclusions.

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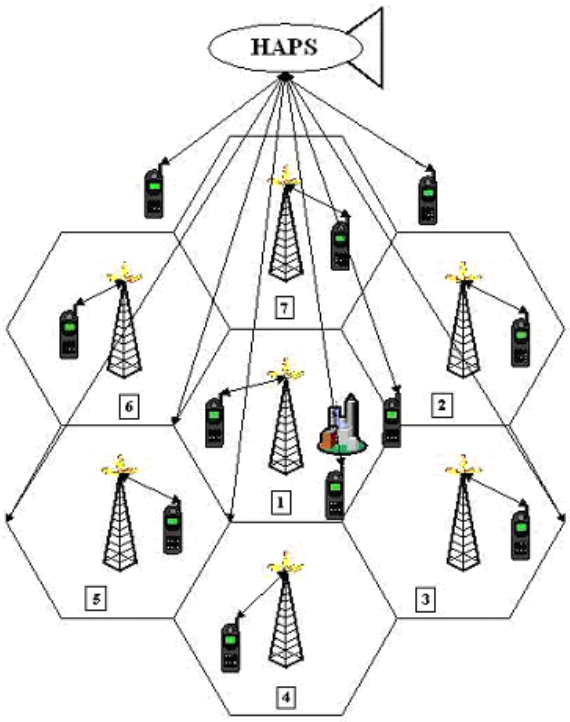


Fig.1: *The conceptual joint system model.*

2. HYSTERESIS METHOD

A traditional handoff algorithm which is known as the Hysteresis method uses relative signal strength as a main component in the handoff decision process. In Fig. 2, the mobile is moving from the serving BS to another BS. To have an ongoing call, we need to handoff when the relative signal strength received from the target base rises above the Hysteresis margin dB. It corresponds to the distance at point C. For GSM system, handoff from one cell to another cell is decided when [1]

$$RSS_AVG = RSS_AVG_T - RSS_AVG_S \geq h$$

where RSS_AVG is the difference between the averaged received signal strength from the target cell (RSS_AVG_T) and from the serving cell (RSS_AVG_S). However, the smaller h is, the more frequent the unnecessary handoffs would happen. In this case, it might result in repeated handoffs between the two BSs caused by rapid fluctuations in the received signal. It is so-called the ping-pong effect. On the other hand, having the larger $h$can increase the decision delay that might cause call dropping. There is therefore a tradeoff between the number of unnecessary handoffs and decision delay.

Thus, we propose an improved handoff algorithm based on RBF networks that consider mobile location, direction, averaged signal power and cell traffic intensity. Namely, the input to the neural network is not only the received signal strengths, but it is also the direction of mobile moving to the target cell and the traffic intensities of the neighbor cells. It makes the proposed approach capable of reducing the handoff rates and blocking rates as well as dropping rates compared to the conventional Hysteresis method.

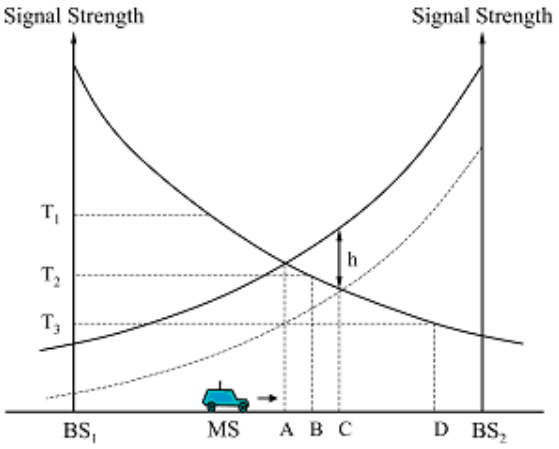


Fig.2: *Location of a handoff decision in Hysteresis*

3. RADIAL-BASIS FUNCTION NETWORK

In this section, we apply radial-basis functions (RBF) to the neural network as depicted in Fig. 3 for making handoff decisions. The basic structure consists of three layers; input layer, hidden layer and output layer [7]. The number of nodes used in the hidden layer is 20. This number was found after training the network and it made the errors converged. The Gaussian functions fully connect them to the five input nodes $\mathbf{x} = [x_1, \ldots, x_5]$. The output node is a linearly weighted sum of the hidden unit outputs. The reason that we have two outputs is to reduce number of iterations of learning process. The outputs decide whether the system needs a handoff or not. If y_1 and y_2 are 00, no handoff will be performed. If y_1 and y_2 are 11, then the system will handoff the mobile to the chosen base station.

The network is trained by the following procedure to find the weight vector,$\mathbf{w}_{k1}(n) = [w_k(n), \ldots, w_{k20}(n)]$, starting at the iteration $n = 0$.

1. Initialize the center value $\mu_{ji}(0)$ of the ith input node and the jth hidden node.

2. Initialize the span value $\sigma_j(0)$ of the jth hidden node.

3. Initialize the weight vector $\mathbf{w}_k(0)$. Note that 1-3 are initialized using a uniform distribution between [0, 1] except $w_{11}(0) = w_{21}(0) = 1$.

4. Calculate the output of the hidden layer:
$z_j = exp\big(-\|x_i - \mu_{ij}(n)\|^2/2\sigma_j^2(n)\big)$

5. Calculate the output:
$y_k = R\Big[\sum_{j=i}^{M} w_{kj}(n) z_j\Big], \quad k = 1, 2; M = 20$
where $R[\bullet]$ is a round operation.

6. Calculate the error: $e_k = d_k - y_k$ where $d_k \in \{0, 1\}$ is the desired pattern.

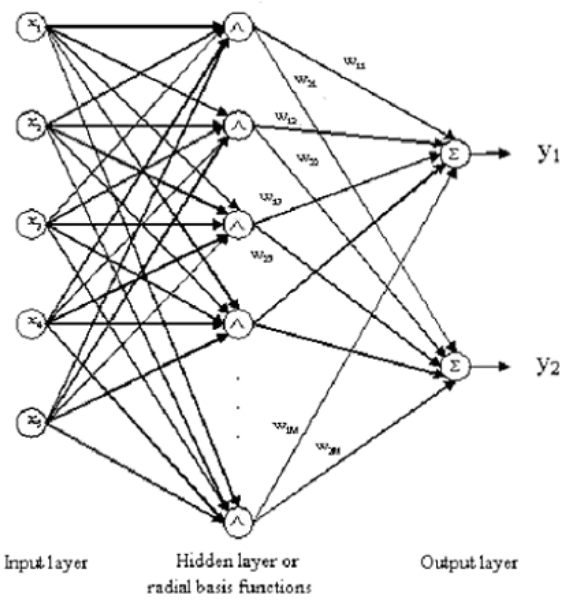


Fig.3: *Radial-basis function network*

7. Update the weight: $w_{kj}(n+1) = w_{kj}(n) + \eta_w e_k z_j$ where η_w is the learning rate of weight.

8. Update the center momentum:
$\mu_{ij}(n+1) = \mu_{ij}(n) + \eta_\mu \frac{z_j}{\sigma_j}(x_i - \mu_{ji}(n)) \sum e_k w_{kj}(n)$
where η_μ is the learning rate of center.

9. Update the span:
$\sigma_j(n+1) = \sigma_j(n) - \frac{2\eta_\sigma z_j}{\sigma_j(n)} \ln z_j \sum e_k w_{kj}(n)$
where η_σ is the learning rate of span.

10. Repeat steps 4-9 until the mean square error converges less than a small fixed number.

The inputs to the neural network are listed below.

1. x_1 is the direction of a user. Under an assumption that the mobile travels in unchanged direction, it can be estimated using the MUSIC algorithm which will be described in the next section.

2. x_2 is the signal strength of mobile received from the serving BS.

3. x_3 is the signal strength of mobile received from the target BS. The received signal strength (RSS) is leveled as:
low: $-96 < RSS \geq -87$dBm and high: $RSS > -87$ dBm.

4. x_4 is the traffic intensity (TI) of the serving BS.

5. x_5 is the TI of the target BS. The TI is leveled as

low: $TI < 0.6591$ Erlangs/Channel

medium: $0.6591 \geq TI \geq 0.76$ Erlangs/ Channel

high: $TI > 0.76$ Erlangs/Channel

Note that, we use the GOS (Grade of Service) in the range 2-5% and the system has 20 channels to design the TI levels. From the Erlang B table [10], TI equals to 13.1816 when the GOS is 2% and equal to 15.2 for 5%. Thus, the traffics per channel are equal to 0.6591 and 0.76, respectively.

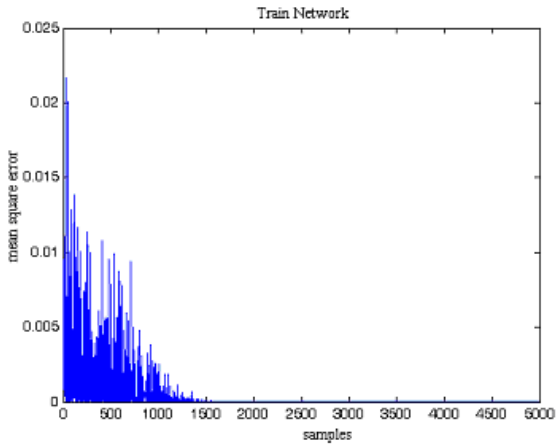


Fig.4: *Mean square errors in the trained network*

The proposed handoff decision by the neural network depends on both RSSs and TIs of the serving (S) and target (T) BSs. We consider four cases as ordered in Table 1 which are

1) Both the RSSs from the serving and target BSs are low.

2) The RSS from the serving BS is low while the RSS of the target BS is high.

3) Both the RSSs from the serving and target BSs are high.

4) The RSS from the serving BS is high while the RSS from the target BS is low.

In each case, there are different handoff decisions (HO: handoff or NOHO: no-handoff) regarding to the levels of traffic intensity.

Consider a particular case that the RSS from the serving cell is low and the RSS of target cell is high and their both intensities are high. In this severe situation, the neural network decides definitely to handoff. If the target cell chosen by using has no channels available, we will consider among the adjacent cells and HAPS. The cell that has the highest RSS will be the cell to where mobile is handoffed.

In our implementation, 5000 samples are used for training the network and 500 samples are used to test the network. For training, it is found that the errors converge to zero as shown in Fig. 4. Figure 5 illustrates the good performance of the network for testing data.

4. BACKPROPAGATION NEURAL NETWORKS

The backpropagation (BP) structure shown in Fig. 6 has the same number of nodes in each layer as Fig. 3. Let $\mathbf{x_p} = (x_{p1}, x_{p2}, x_{p3}, \ldots, x_{pN})$ be a network input vector for N input nodes of pth the learning sequence. In the nth iteration, $v_{ij}(n)$ is the weight element connecting the ith input node and the jth hidden node which is connected to the kth output node

Table 1: *Handoff decision*

RSS from Target BS : Low	RSS from Serving BS : Low			
	TI : S / TI : T	Low	Medium	High
	Low	NOHO	HO	HO
	Medium	NOHO	NOHO	HO
	High	NOHO	NOHO	NOHO

RSS from Target BS : High	RSS from Serving BS: Low			
	TI : S / TI : T	Low	Medium	High
	Low	HO	HO	HO
	Medium	HO	HO	HO
	High	NOHO	NOHO	HO

RSS from Target BS: High	RSS from Serving BS: High			
	TI : S / TI : T	Low	Medium	High
	Low	NOHO	HO	HO
	Medium	NOHO	NOHO	HO
	High	NOHO	NOHO	NOHO

RSS from Target BS: Low	RSS from Serving BS: High			
	TI : S / TI : T	Low	Medium	High
	Low	NOHO	NOHO	HO
	Medium	NOHO	NOHO	HO
	High	NOHO	NOHO	NOHO

by $w_{jk}(n)$. In the forward propagation, the input of the jth hidden node is

$$net^h_{pj} = \sum_{i=1}^{N} v_{ij}(n) x_{pi}$$

where $v_{ij}(0) = 1$. The output of the jth hidden nodes for $j = 1, \ldots, M$ can be obtained as

$$z_{pj} = \varphi(net^h_{pj})$$

where $\varphi(x) = \frac{1}{1+e^{-x}}$ is the sigmoid function. Once the output of the hidden node z_{pj} is found, we can

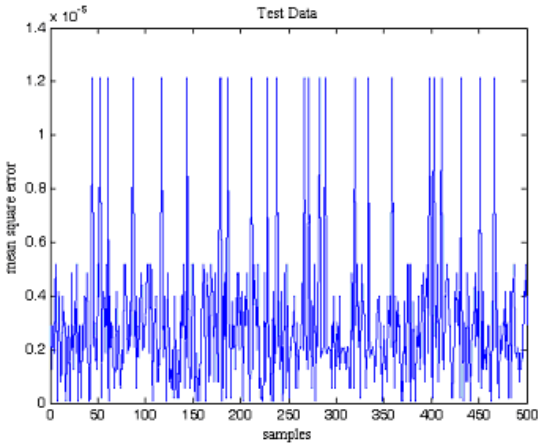


Fig.5: *Mean square errors in the tested network*

compute the input of the output layer as

$$net^0_{pk} = \sum_{j=1}^{M} w_{kj}(n) z_{pj}$$

where $w_{kj}(0) = 1$. Note that the weight vectors are initialized uniformly distributed between [0, 1]. The output of the kth output node is a function as

$$y_{pk} = R[\varphi(net^0_{pk})]$$

In backward propagation, the procedure is started from the output layer to the input layer to adjust the output and hidden weights given as follows:

$$\delta_{pk} = \varphi'(net^0_{pk})(d_k - y_{pk})$$

where $\varphi'(x)$ represents the differential. Next, update the weight elements between the hidden and the output layers

$$\Delta w_{kj}(n+1) = \eta \delta_{pk} z_{pi} + \mu \Delta w_{kj}(n).$$

where η is the learning rate and μ is the momentum. It is used to update the weight in the next iteration by adding

$$w_{kj}(n+1) = w_{kj}(n) + \Delta w_{kj}(n+1).$$

Likewise, the weights connecting the hidden and input layers are updated by using the following equations.

$$\delta_{pj} = \varphi'(net^h_{pj}) \sum_k \delta_{pk} w_{kj}(n+1)$$

$$\Delta v_{ji}(n+1) = \eta \delta_{pj} x_{pj} + \mu \Delta v_{ji}(n)$$

$$v_{ji}(n+1) = v_{ji}(n) + \Delta v_{ji}(n+1).$$

After updating $v_{ji}(n+1)$ and $w_{kj}(n+1)$, the forward propagation is repeated until the errors converge.

10000 samples are used for training and 500 samples are used for testing. The convergence of error is shown in Fig. 7. Figure 8 shows the mean square error resulting from testing the network.

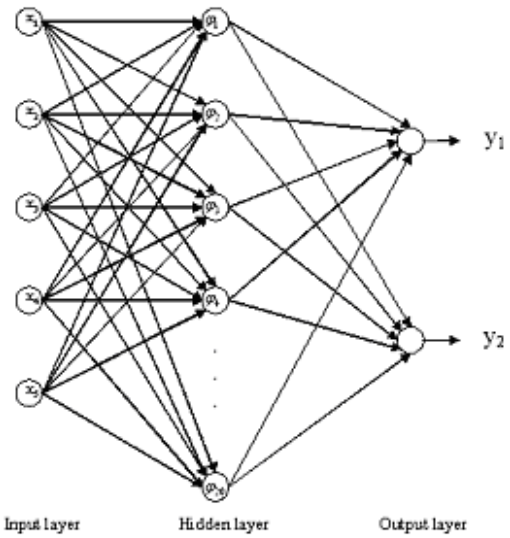


Fig.6: *Back-propagation neural network.*

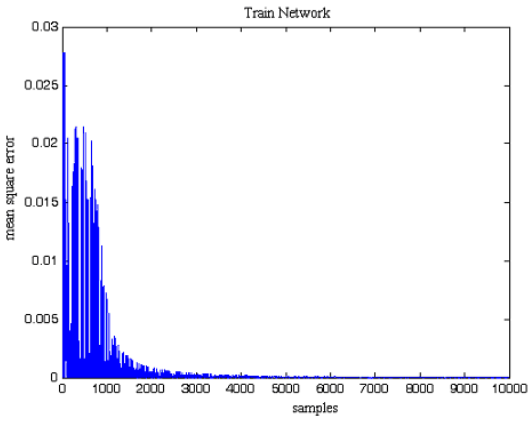


Fig.7: *Mean square errors in the trained network*

5. THE MUSIC ALGORITHM

Consider the received signals impinging on L uniform linear array antennas which consist of the M source signals corrupted by noise in the form of [9]:

$$\mathbf{X}(t) = \sum_{m=1}^{M} \mathbf{a}(\theta_\mathbf{m})\mathbf{s_m}(t) + \eta(t) = \mathbf{A}(\mathbf{\Theta})\mathbf{s}(t) + \eta(t)$$

where $\mathbf{A}(\mathbf{\Theta}) = [\mathbf{a}(\theta_\mathbf{1}), \ldots, \mathbf{a}(\theta_\mathbf{M})]$ is an $L \times M$ steering matrix whose column is a steering vector $\mathbf{a}(\theta_\mathbf{m})$ of the signal $s_m(t)$ having an unknown angle of arrival θ_m. $\mathbf{\Theta} = [\theta_\mathbf{1}, \ldots, \theta_\mathbf{M}]$ is an angle of arrival (AOA) vector of the M source signal vector denoted as $\mathbf{s}(t) = [\mathbf{s_1}(t), \ldots, s_M(t)]^T$ with an assumption that $M < L$. The steering vector can be written as:

$$\mathbf{a}(\theta_\mathbf{m}) = [\mathbf{1e}^{-\mathbf{j2}\pi\frac{\mathbf{d}}{\lambda}\mathbf{sin}\theta_\mathbf{m}} \ldots \mathbf{e}^{-\mathbf{j(L-1)}2\pi\frac{\mathbf{d}}{\lambda}\mathbf{sin}\theta_\mathbf{m}}]^\mathbf{T}$$

where d/λ is the ratio of inter-element space of array to the signal wavelength. $\eta(t) = [\eta_1(t), \ldots, \eta_L(t)]^T$

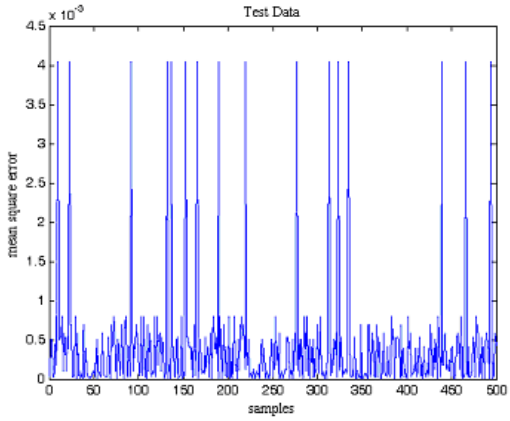


Fig.8: *Mean square errors in the tested network*

is a noise vector which is temporally and spatially white correlated.

The correlation matrix $\mathbf{R}$ of the array signal output can be written as

$$\mathbf{R} = \mathbf{E}[\mathbf{x}(t)\mathbf{x^H}(t)].$$

An expression of $\mathbf{R}$ in terms of eigenvalues and their corresponding eigenvectors is given as

$$\mathbf{R} = \mathbf{U\Lambda U^H}$$

where $\mathbf{U} = [\mathbf{u_1}, \ldots, \mathbf{u_L}]_{\mathbf{L}\times\mathbf{L}}$ is a unitary matrix and Λ is a diagonal matrix of real eigenvalues $\lambda_i(\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_L > 0)$ corresponding to the eigenvector u_i. Then, we decompose the correlation matrix $\mathbf{R}$ into a signal subspace: $\mathbf{S} = [\mathbf{u_1}, \ldots, \mathbf{u_M}]$ and a noise subspace: $\mathbf{N} = [\mathbf{u_{M+1}}, \ldots, \mathbf{u_L}]$. Thus, the matrix $\mathbf{R}$ can be rewritten as a sum of the two subspaces:

$$\mathbf{R} = \mathbf{S\Lambda_s S^H} + \mathbf{N\Lambda_n N^H}.$$

As a result of orthogonality of the steering vectors to the noise subspace, the MUSIC algorithm estimates the AOA vector of the signals by finding M peaks of

$$P_{MUSIC}(\theta) = |\mathbf{a^H}(\theta)\mathbf{N}|^{-\mathbf{2}}.$$

In practice, we cannot find the correlation matrix $\mathbf{R}$, thus we use the sample-average estimate for N snapshots as

$$\hat{\mathbf{R}} = \frac{1}{N}\sum_{t=0}^{N-1} \mathbf{x}(t)\mathbf{x^H}(t).$$

Figure 9 shows an AOA estimate of the ten source signals corresponding to the ten peaks of MUSIC. The FSK signal is used and defined as [13]

$$s(t) = \sqrt{\frac{2E_s}{T}} cos[\omega_c t + \theta(t)]$$

where ω_c is the carrier frequency, $\theta(t)$ the phase of the signal, E_s is the symbol energy and T is the symbol

period. In simulation, we assign T equal to 0.02 secs, ω_1 and ω_2 are equal to 100 and 200 Hz, $E_s = 1$ dB and $\theta(t)$ is generated using the uniform distribution in the range of $0 - 2\pi$. The number of antennas is used 30.

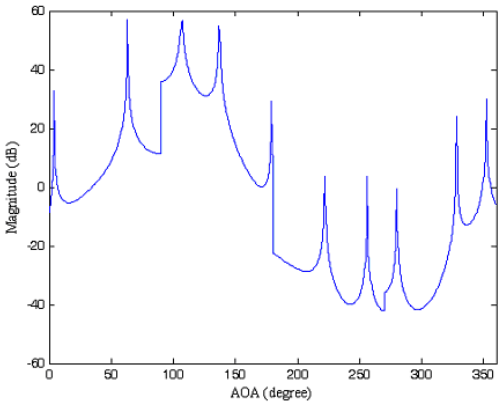


Fig.9: *AOA estimates at* 4°, 63°, 107°, 137°, 179°, 222°, 256°, 280°, 328°, 352°.

6. TIMING ADVANCE

In the proposed handoff method, we need to know the mobile locations (d) to check if the mobiles are in an obstacle position or in the 3-cell junction. Positioning a mobile station is divided into two categories: network-based and handset-based positioning. We use one of handset-based technique called the timing advance (TA) [11-12] to determine where the mobile is.

TA is a measurement of the time required for the signal to travel from the MS to the BS. The concept of TA is used in order that the time-frames from each MS arrive at the correct time slot when received by the serving BS. By checking the position of the training sequence transmitted from the MS on the uplink, the BS can calculate the TA value and send it back to the MS in the downlink. Then, it enables the mobiles to know when to send the frame so that the signals arrive at the BS in synchronism. For example, the value of TA in GSM system is an integer in a range of 0-63 according to 0 to 233 μs. It defines 0 is to be no TA and 63 to be the maximum TA. Each TA number corresponds to different 550 m. radial distance. It is reported to the MS every 480 ms during the connection.

In this paper, the TA is set in the range of 0 to 7 according to 0 to 7 μs. These values are coded by 3 bits. One bit of the TA represents a time difference of (7 μs)/7= 1 μs of the signal BS-MS-BS. The distance per bit of TA is $d_b = (3 \times 10^8 \times 1\mu s)/2 = 150$ m/TA. The MS-BS distance is a function of the integer TA as

$$d(TA) = TA \times d_b$$

where $TA = R[\nu\Delta t/150]$ depending on the velocity of mobile ν and a time interval of measurement Δt.

7. NEURAL NETWORK HANDOFF ALGORITHM

Figure 10 is a flow chart illustrating the proposed handoff algorithm. The capacity of the system is more efficient with the goodness of HAPS. First, we check if the mobile is staying in the obstacle position or even at the corners of 3-cell junction by monitoring if the mobile distance is $d < 800$ or $d \geq 1000$ and if the RSS is less than the threshold. The threshold of the received signal strength (RSS_drop) is at -82 dBm before call is dropped. If the mobile has the signal strength received from the serving BS less than -82 dBm, a handoff from the serving terrestrial BS to HAPS-BS is necessary to improve the call quality. If $800 \leq d \leq 1000$, we measure the signal strength of the mobile received from the current and adjacent cells every 0.5 sec. In order to decrease variation of the signals, an average of 10 data is calculated. After obtaining the averages of signal strength of current and neighbor cells of x_1, we can determine which cells are most likely to be the target base stations (BST(i)) by checking that the mobile receives the signal power from which base stations radiate greater than -96 dBm (RSS_low_TH). Included the knowledge of the mobile directions obtained by the MUSIC algorithm and traffic intensities of the chosen BSs, the trained neural network then makes a decision if the mobile needs a handoff or not. In the case of deciding to handoff, the algorithm will assign either case (1) to which base station the mobile will be handoffed or (2) request a channel from HAPS. Note that HAPS gives higher priority to handoff-calls than new-calls such that probabilities of blocking calls are low.

8. MOBILE RADIO PROPAGATION MODEL

8.1 Hata Model

The path loss in Hata model is expressed as [10]
$P_L = 69.55 + 26.16\log f_c - 13.82\log h_b - a(h) + (44.9 - 6.55\log h_m)\log d$
where $a(h) = 3.26(\log 11.75 h_m)^2 - 4.97$ for a carrier frequency $f_c \geq 300$ MHz, h_b is the height of antenna at base station (m) h_m is the height of antenna at mobile (m), d is the distance between BS and MS (km). Accordingly, the strength of signal at a mobile is

$$RSS = P_0 - P_L$$

where P_0 denotes the power radiated from the BS.

8.2 Shadow-Rayleigh Fading

In mobile radio channels, the Rayleigh distribution [4,10] is commonly used to model a flat fading channel. This small-scale fading describes the statistical time varying nature of the received envelope. It is

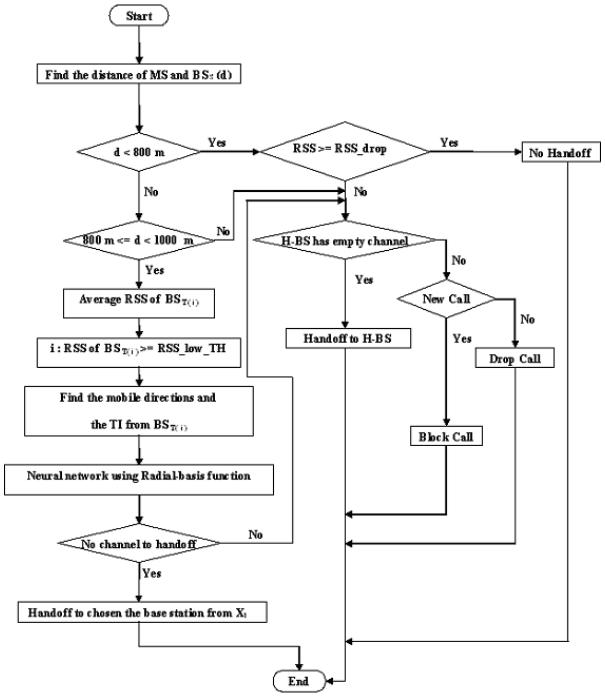


Fig.10: *Handoff algorithm using neural network.*

well known that the sum of signals with the random amplitudes and phases by no direct path between a MS and a BS obeys a Rayleigh distribution. The envelope of the received signal strength r_k at a MS-BS distance d has a probability density function given by

$$p(r_k) = \frac{r_k}{p_k}\exp\Big(-\frac{r_k^2}{2p_k}\Big), \;\; 0 \le r_k \le \infty$$

where p_k is the time-average power of the received signal strength before envelope detection. The mean value of the Rayleigh distribution is $E[r_k] = \sqrt{\frac{\pi p_k}{2}}$. The received signal power with an arbitrary transmitter-receiver separation distance is modeled as using the path loss exponent from 2 to 6.

The shadow fading process is a Gaussian distributed random process measured in decibels (dB). Under lognormal shadow fading, the autocorrelation is given as

$$R_L(d) = \sigma_L^2 \exp\Big(-\frac{|d|}{d_0}\Big)$$

where σ_L^2 and d_0 are the variance and correlation length of shadow fading, respectively. Accordingly, the power spectral density $S_L(v)$ can be found as

$$S_L(v) = \frac{2d_0\sigma_L^2}{1+(2\pi v d_0)^2}$$

where v is a spatial frequency. For a total traveled distance D, the shadow fading process can be written as

$$L(d) = \sum_{j=-J}^{J} \sqrt{\frac{2}{BD}S_L\Big(\frac{j}{D}\Big)}\cos\Big(\frac{2\pi j d}{D}+\phi_j\Big)$$

where

$$B = \frac{1}{\sigma_L^2 D}\sum_{j=-J}^{J} S_L\Big(\frac{j}{D}\Big), \;\; J = Dv_m.$$

Let v_m be the maximum spatial frequency and ϕ_j be an identically independent uniform random process in range of $0-2\pi$. The discrete process L_k is obtained by sampling the process at $d = d_k$.

The received signal strength (RSS in dB) in the presence of Rayleigh fading is in the form of

$$s_k = 20\log_{10} r_k$$

where r_k is a Rayleigh distribution random variable with parameter $p_k = (1/2)10^{m_k/10}$. The power can be calculated in dB as

$$m_k = P_0 - 10n\log_{10} d_k + L_k.$$

The mean and variance of are given as

$$\begin{aligned} \bar{s}_k &= 10\log_{10}(2p_k) - \frac{10\gamma}{\ln 10} \\ &= m_k - \frac{10\gamma}{\ln 10} \\ \sigma_{s_k}^2 &= \frac{50\pi^2}{3\ln^2 10} \end{aligned}$$

where $\gamma \approx 0.577216$ is Euler's Gamma.

9. SIMULATION RESULTS

The performance of the handoff algorithm using the RBF and BP neural networks is illustrated in comparison to the Hysteresis method. Simulations are done for the system pictorially shown in Fig. 1. 200 trials are run at a fixed traffic intensity and at a fixed mean arrival time of the serving BS. The carrier frequency is $f_c = 1800$ MHz. The antenna height is 60 m at BS and 1.5 m. at MS. The parameters of cell environment and the statistical parameters of mobile are listed in Table 2-3. To evaluate the proposed method and Hysteresis rule, the handoff rate (H/T_h), blocking rate (B/T) and dropping rate (D/T_h) are computed at the serving BS (cell no. 1). H represents the number of successful handoffed calls; Th represents the total number of calls decided to handoff; B represents the number of blocked calls; T_h represents the total number of new calls; D represents the number of dropped calls. There are two cases in each channel model. First, we vary the traffic intensities in the range of 0.65-0.77 Erlang/channel for acceptable GOS 2-5% of the cell No. 1 at the mean arrival time equal to 1 min. In the other case, we vary the mean arrival time from 3-5 mins.

In Hata path loss propagation, Figs. 11-17 show that handoff algorithm using the RBF network outperforms that using BP network and Hysteresis. Handoff rate, blocking rate and dropping rate resulting from using RBF network are lowest in any case

of traffic intensities and mean arrival times as plotted in Figs. 11-16. Since we consider the signal strength, position, direction and traffic intensity, the unnecessary handoffs is reduced as seen in Figs. 11 and 14. The resulting blocking rates in Figs. 12 and 15 have also low probabilities. As a result of the reduced unnecessary handoffs and HAPS, there are more available channels for new calls to access the system. We also achieved the reduction of dropping rates as shown in Figs. 13 and 16 because once the handoff rate decreases, chance of dropping is also less. Fig. 17 depicts the number of handoffs occurring at the difference between the signal power of mobile receiving from the current BS and the minimum received signal strength (-96 dBm) that the call will dropped. In fact, the less power difference, the more number of handoffs. Using the proposed algorithm, low number of handoffs at low power difference is achieved. For Hysteresis rule, the power difference should higher than 6 dB to avoid call-dropping.

In shadow-Rayleigh fading, we assign the Rayleigh fading parameters in Table 3. Similarly, the results obtained from the proposed method using the RBF network are the best among using BP network and using Hysteresis as shown in Figs.18-24.

10. CONCLUSION

We propose an effective handoff algorithm based on RBF network in high altitude platform station cellular systems. HAPS system can provide services to the users staying at the corner of cells or at covered area influenced by shadowing. Besides the averaged received signal strength, user position and direction and traffic intensity are inputs of the neural networks. We use the RBF and BP to construct the networks. The MUSIC technique in the antenna array provides us an estimate of user direction. Timing advance used in the GSM system is applied to this system for mobile positioning. Consequently, handoff rate, blocking rate and dropping rate decrease compared with the traditional Hysteresis algorithm. The simulations are performed in large scale channel propagation which is the model of Hata path loss, and small scale channel propagation in the process of Rayleigh fading. Besides the lower rates, we achieve low difference between the mobile signal power from the target BS and the minimum required power before dropping calls at a few needed numbers of handoffs.

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Table 2: *Cell Structure Parameters*

Parameters	value
cell radius	1000 m
number of channels per cell	20 channels
number of channels in HAPS	20 channels
low threshold of received signal strength (RSS_low_TH)	-96 dB
received signal strength for call dropping (RSS_drop)	-82 dB
Power Transmitted p_0	40 dBm
Hysteresis level (h)	6 dB

Table 3: *Mobile Parameters*

variable	distribution	interval
initial position in the horizontal (m)	Uniform	[-10,10]
initial position in the vertical (m)	Uniform	[-8,8]
number of users in each cell	Uniform	[100,400]
mean arrival time of new call	Poisson	1 min
direction of mobile	Uniform	[0,2π]
average time per call	Exponential	120 sec
average mobile speed	Normal	N(60,10) km/hr

Table 4: *Rayleigh Fading Parameters*

Parameters	value
Minimum Base Station Separation	1600 m
Exponent of Distance Dependence (n)	4
Standard Deviation of Shadow Fading σ_L	8 dB
Correlation Length of Shadow Fading d_0	20 m
Number of terms for Shadow Fading Process	321
Maximum Spatial Frequency	0.1

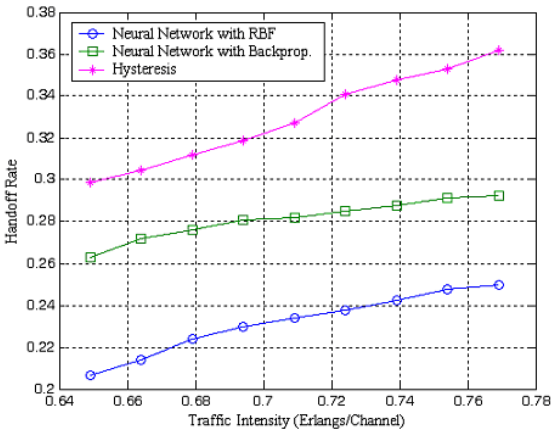


Fig.11: *Handoff rate versus traffic intensity in Hata*

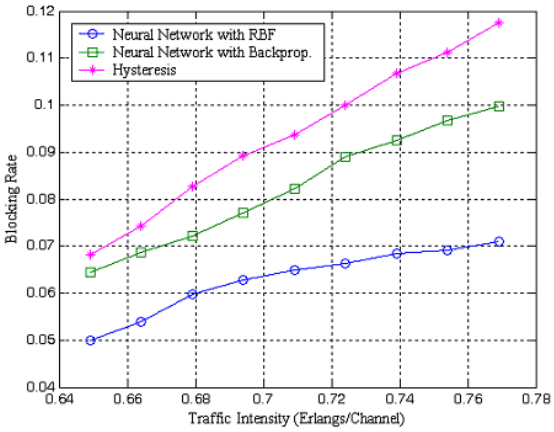


Fig.12: *Blocking rate versus traffic intensity in Hata*

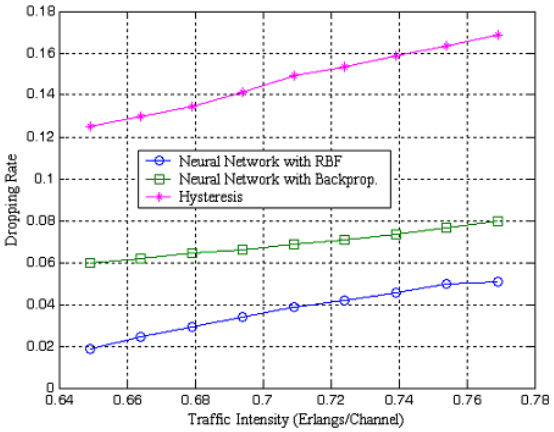


Fig.13: *Dropping rate versus traffic intensity in Hata*

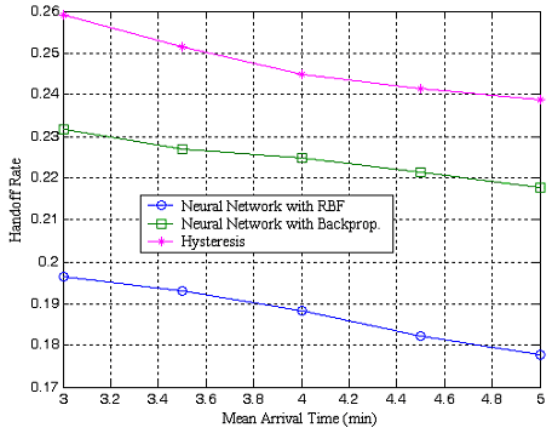


Fig.14: *Handoff rate versus mean arrival time in Hata*

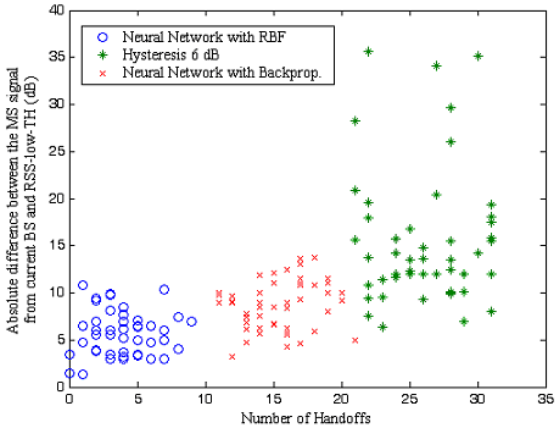


Fig.17: *Signal difference versus number of handoffs in Hata*

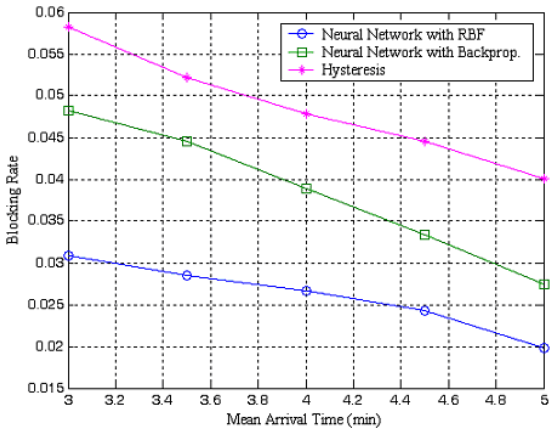


Fig.15: *Blocking rate versus mean arrival time in Hata*

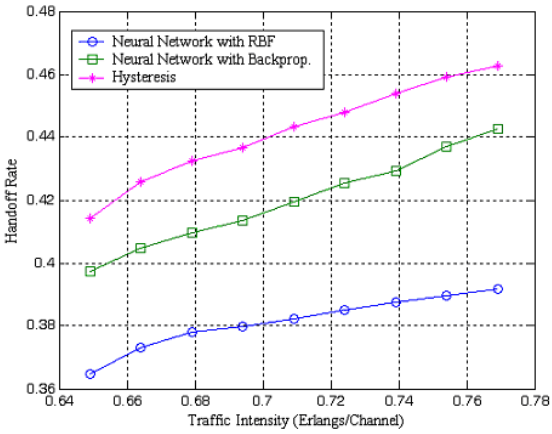


Fig.18: *Handoff rate versus traffic intensity in Rayleigh*

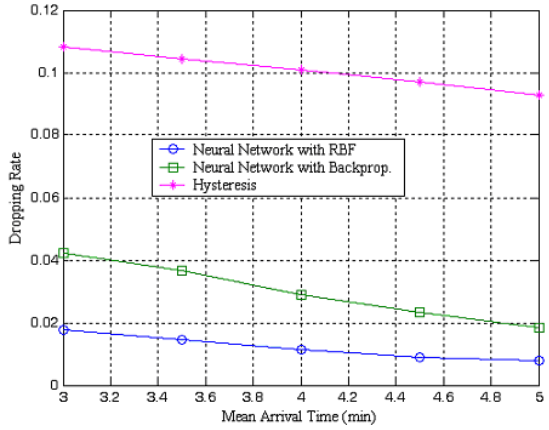


Fig.16: *Dropping rate versus mean arrival time in Hata*

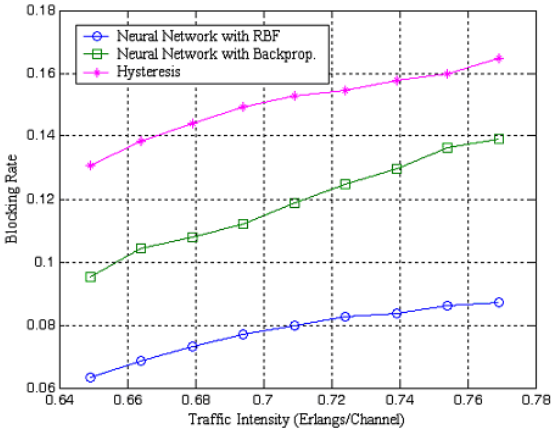


Fig.19: *Blocking rate versus traffic intensity in Rayleigh*

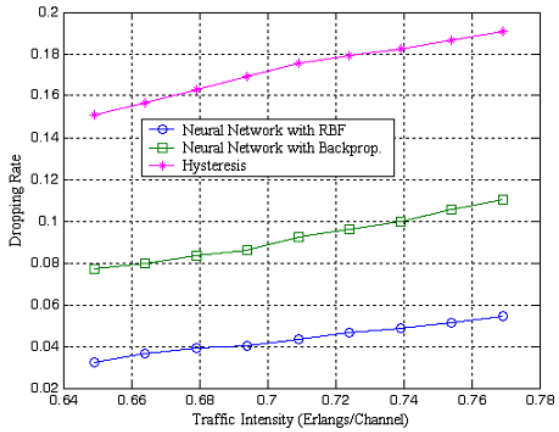


Fig.20: *Dropping rate versus traffic intensity in Rayleigh*

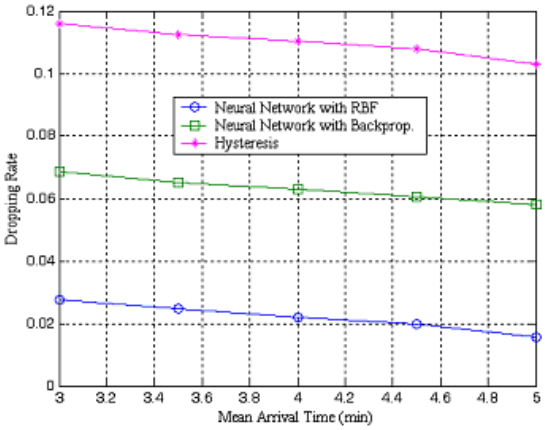


Fig.23: *Dropping rate versus mean arrival time in Rayleigh*

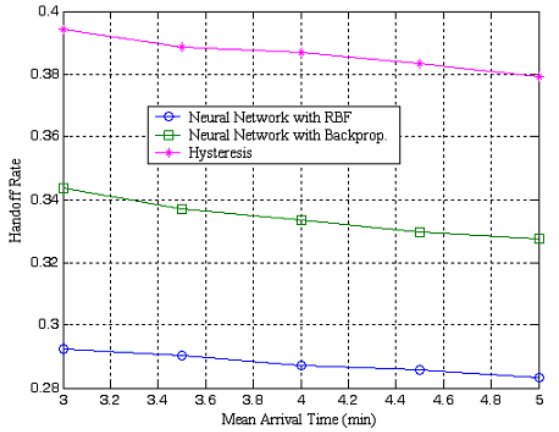


Fig.21: *Handoff rate versus mean arrival time in Rayleigh*

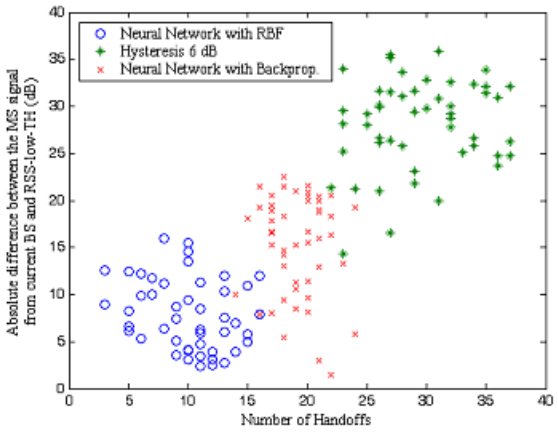


Fig.24: *Signal difference versus number of handoffs in Rayleigh*

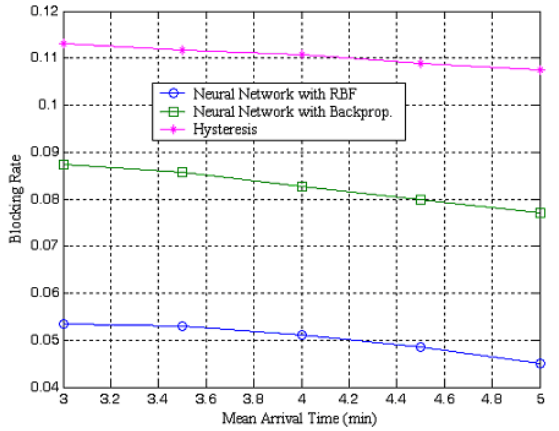


Fig.22: *Blocking rate versus mean arrival time in Rayleigh*